From Perception to Punchline:
Empowering VLM with the Art of In-the-wild Memes

HUMOR

Hierarchical Understanding and Meme Optimization

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1 What makes a caption a meme?

Meme generation connects visual relationships to a situation and a punchline. Naming objects is not enough; the same template can express many different intentions.

Input: a template plus user intent (emotion, theme, style or keywords).
Output: captions assigned to the template's text regions.

One template, three different outcomes



Human reference | **Single-path CoT** | **HUMOR-CoT**

Fig. 16 examples. Single-path reasoning repeats anticipation; the reference supplies a concrete obstacle. HUMOR learns a context-dependent delayed-reward situation.

2 Why compare within a template?

A global "funniness score" assumes that unrelated jokes share one scale. We instead learn relative quality inside a coherent group G, usually one template.

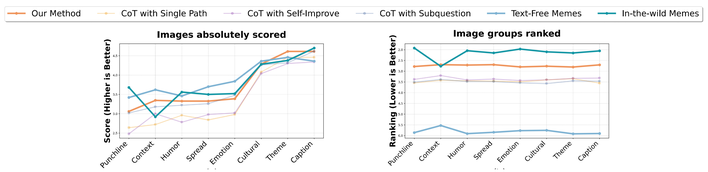


Fig. 4. Absolute VLM scores (left) poorly separate real memes from blank images; group-wise ranks (right) provide clearer distinctions. Lower rank is better.

Independent check: 9 annotators, 5 groups, 30 memes. Gemini–human agreement: Spearman 0.72; Kendall τ = 0.63.

Data: 3,345 train / 368 test instances. Teacher: Doubao-1.5-Vision-Pro. Human ratings and MaxDiff rankings are separate from constructed training pairs.

3 Explore contexts, then ground a path

Direct caption supervision hides the associations behind a joke. HUMOR builds traces that preserve several plausible contexts before grounding one path in a real meme.



Template interpretation

Attention shifts from an existing commitment to a new attraction.

These context branches illustrate the reusable relation; they are not scored outputs.

MULTI-CONTEXT EXPLORATION

- Technology → a new phone
- Work → a new opportunity
- Daily life → a tempting purchase

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A real caption supplies the actual context; bind its roles to regions and realize the punchline.

AT INFERENCE

Template + user intent → hypothesize contexts → select a path → positioned caption. No reference caption is used.

4 Learn preferences, not global scores

Training pairs share a template. Five constructed difficulty tiers teach a VLM reward model to distinguish disrupted meaning, weak humor and plausible alternatives.

Wrong text
unrelated caption

Wrong location
misplaced roles

Boring
mundane statement

Detailed boring
small edits weaken the joke

Generated
harder model-written alternatives

Pairwise probability $p(i > j \mid G) = \sigma[s(i) - s(j)]$

Expected Borda Count $EBC(i) = \Sigma\, p(i > j \mid G),\ j \neq i$

Binary cross-entropy trains the pairwise comparator. Expected wins induce a group-local preference distribution q. The policy learns to favor higher-ranked captions while staying near its CoT reference.

Policy loss = $-E[\Sigma\, q \log \pi(c \mid I)] + \beta\, E[KL(\pi \,||\, \pi ref)]$

5 Generation quality and ablations

30 native-English raters; 3–5 memes per method per rater. Human scores are means on a 0–5 scale (↑). All fine-tuned variants use the 7B backbone.

Model	Hum.	Read.	Rel.	Orig.	Dist.
Qwen2.5-7B	2.39	3.35	2.91	2.57	0.564
Qwen2.5-32B	2.54	3.52	3.09	2.76	0.566
InternVL3-8B	2.39	2.79	3.04	2.79	0.564
GLM-4.1V-9B-Thinking	1.73	2.62	2.75	2.71	0.572
Keye-VL-8B-preview	2.35	3.19	2.99	2.71	0.580
GPT-4o	2.70	2.99	3.21	**2.97**	0.552
Gemini-2.5-flash	2.81	3.29	3.25	2.88	0.561
HUMOR-CoT	2.68	**3.70**	3.50	2.90	0.590
CoT: Single Path	1.87	2.79	2.68	2.45	0.570
CoT: Self-Improve	2.38	3.68	3.00	2.65	0.578
CoT: Subquestion	1.85	3.32	2.58	2.47	**0.597**
HUMOR-RL (preview)	**2.83**	3.67	**3.55**	2.79	0.588

Table 1. All 12 models/variants. Dist. = context-swap distance, a diversity proxy. Human/AI discriminability is omitted. Bold: column best.

CoT improves humor 2.39 → 2.68; single-path CoT falls to 1.87. RL reaches 2.83 in preliminary results, with lower originality than CoT (2.79 vs. 2.90).

6 Does the reward agree with people?

Kendall τ against human MaxDiff rankings: 100 participants, 5 templates, 10–15 candidates per group.

Reward backbone / state	G1	G2	G3	G4	G5
Qwen2.5-7B · base	0.16	0.28	0.47	-0.10	0.29
Qwen2.5-7B · trained	0.47	0.56	0.42	0.14	0.47
Qwen2.5-32B · base	0.16	0.16	-0.02	0.14	0.29
Qwen2.5-32B · trained	0.29	0.47	0.07	0.30	0.42
Keye-VL-8B · base	0.05	0.09	0.16	0.29	0.16
Keye-VL-8B · trained	**0.78**	**0.77**	**0.78**	**0.78**	**0.78**

Table 2. Keye reward training consistently reaches τ ≈ 0.78. Improvements are less consistent on Qwen backbones; ranking quality depends on the reward model.

Pair construction is synthetic/model-assisted; the evaluation uses human rankings.

7 What the analysis guarantees

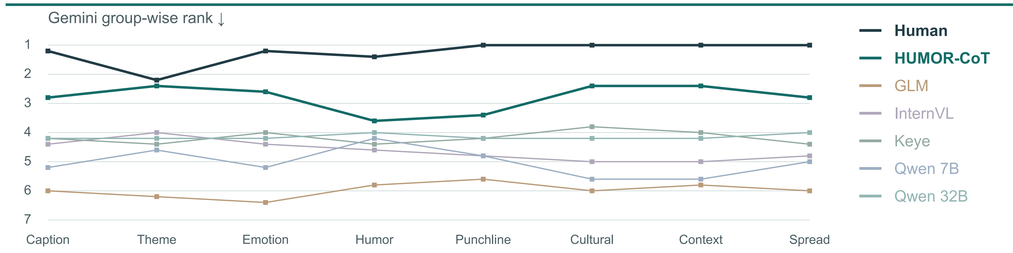
Exploration $E[h] \geq 1 - (1 - \alpha)\delta$

If promising paths retain mass α and the remaining quality gap is at most δ, expected normalized humor has this lower bound. More paths alone do not establish the assumptions.

KL stability $H(\pi) \geq H(ref) - \sqrt{(\Delta KL / 2)}$

For bounded humor h in [0,1], KL limits possible degradation. Preference consistency additionally assumes a monotone observation model; neither result guarantees monotonic gains.

8 Generalization to 20 unseen templates



Second to human memes across all eight dimensions.

Scope: English meme culture; limited human samples; ~2–3× output-token cost; concrete keywords can induce literalism.

Paper: arxiv.org/abs/2512.24555 | karroyan13@gmail.com